Solution Combustion Synthesis of Metal Nanorods and Nanowires: Mechanistic Foundations and the Limits of One-Dimensional Growth

Abstract: Solution combustion synthesis (SCS) is a rapid, energy-efficient technique widely used for producing oxide nanomaterials. Its extension to one-dimensional (1D) metallic nanostructures—nanorods and nanowires—remains fundamentally challenging due to the inherent conflict between the isotropic, gas-releasing combustion process and the anisotropic growth required for 1D morphology. This paper presents a detailed mechanistic analysis of SCS for metal nanostructure synthesis, with copper as the model system. Thermodynamic considerations based on propellant chemistry and Gibbs free energy minimization are discussed, establishing the critical role of the fuel-to-oxidizer ratio (φ) in determining combustion temperature and product phase. Experimental evidence from $Cu(NO_3)_2$–glycine and $Cu(NO_3)_2$–citric acid systems reveals that metallic Cu formation proceeds through a two-stage pathway: initial oxide formation in the combustion front followed by reduction in the post-combustion zone under fuel-rich conditions. While SCS successfully produces metallic Cu nanoparticles and CuO nanorods, the direct synthesis of metallic Cu nanowires via SCS has not been achieved. The mechanistic origins of this limitation are analyzed, and the conditions under which SCS-derived 1D structures (primarily oxides) can form are discussed.

Keywords: Solution combustion synthesis; Copper nanoparticles; Nanorods; Nanowires; Propellant chemistry; Adiabatic flame temperature

1. Introduction

Solution combustion synthesis (SCS) has emerged as a versatile and industrially relevant technique for nanomaterial production, offering advantages of simplicity, energy efficiency, and molecular-level mixing of reactants . Since its introduction by Patil in 1988 for alumina synthesis, SCS has been applied to over 1,000 types of oxide powders worldwide . The technique relies on a self-sustained exothermic reaction between an oxidizer (typically metal nitrates) and a fuel (e.g., glycine, urea, citric acid), initiated at relatively low temperature and driven by internal chemical energy .

Despite its success for oxide nanomaterials, the application of SCS to **metallic** nanostructures—and particularly to **one-dimensional (1D)** metallic nanorods and nanowires—presents fundamental challenges. The rapid, isotropic nature of the combustion process favors spherical nanoparticle formation, while 1D growth requires sustained anisotropic deposition that is difficult to achieve within the sub-second combustion timeframe. Nevertheless, understanding the mechanistic basis of SCS is essential for evaluating its potential and limitations for 1D metal synthesis.

This paper provides a detailed theoretical and experimental analysis of SCS for metal nanostructure synthesis, using copper as the model system. The thermodynamic framework governing SCS reactions is presented, followed by an examination of the reaction pathways for Cu formation and the conditions under which 1D morphologies can (or cannot) be achieved.

2. Theoretical Foundations of Solution Combustion Synthesis

2.1 Propellant Chemistry and Stoichiometry

The stoichiometry of SCS reactions is conventionally calculated using **propellant chemistry theory**, which assigns elemental valences to determine the oxidizing and reducing powers of reactants . For a combustion reaction, the total oxidizing valence of the metal nitrate must balance the total reducing valence of the fuel.

The oxidizing valence of a metal nitrate is calculated from the sum of elemental valences: for $Cu(NO_3)_2$, Cu = +2, N = 0, O = −2, yielding an oxidizing valence of −10 per formula unit . For glycine ($C_2H_5NO_2$), the reducing valence is +10; for citric acid ($C_6H_8O_7$), it is +18 .

The fuel-to-oxidizer ratio is expressed as φ:

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$$\phi = \frac{\text{actual fuel/oxidizer ratio}}{\text{stoichiometric fuel/oxidizer ratio}}$$

where φ = 1 represents the stoichiometric condition requiring no atmospheric oxygen for complete fuel oxidation. φ > 1 indicates fuel-rich conditions, and φ < 1 indicates fuel-lean conditions .

The general SCS reaction for a metal nitrate–fuel system can be written as :

$$\text{M(NO}_3\text{)}_x + \phi \cdot \nu \cdot \text{fuel} + \left(\frac{x}{2}\right)(\phi - 1)\text{O}_2 \rightarrow \text{MO}_{x/2} + \text{gases}$$

For fuel-rich conditions (φ > 1), the excess fuel generates reducing species (NH_3, H_2, CO) that can reduce metal oxides to metals .

2.2 Thermodynamic Modeling: Adiabatic Flame Temperature

The **adiabatic flame temperature (T_ad)** represents the maximum temperature achievable under adiabatic conditions and is a critical parameter governing product phase, crystallinity, and particle size . T_ad is calculated using the Gibbs free energy minimization principle, simultaneously determining equilibrium product composition and temperature .

For a $Ni(NO_3)_2$–glycine system, thermodynamic calculations show that T_ad increases with φ, reaching a maximum of 2,480 K at φ = 3 with zero water content . This trend reflects the exothermic nature of fuel oxidation: higher fuel content releases more heat, up to the point where excess fuel and incomplete combustion begin to have a counteracting effect.

The effect of water content (from hydrated precursors) is significant: increasing water reduces T_ad due to the enthalpy of vaporization . This has practical implications, as precursor drying time controls residual water content and thus combustion temperature .

For $Cu(NO_3)_2$-based systems, thermodynamic calculations reveal that the combustion temperature and product composition are strongly φ-dependent. At φ = 1 (stoichiometric), the system produces primarily CuO; at φ > 1 (fuel-rich), metallic Cu becomes thermodynamically favored as reducing gases (H_2, CO, NH_3) dominate the product gas composition .

2.3 Reaction Pathway for Metal Formation

The formation of metallic copper via SCS does not occur directly in a single step. Combined X-ray diffraction (XRD) and differential thermal analysis–thermogravimetric analysis (DTA–TGA) studies of the $Cu(NO_3)_2 \cdot 6H_2O$–glycine system have established a **two-stage reaction pathway** :

Stage 1 (Combustion Front): $Cu(NO_3)_2$ decomposes to form oxide phases (Cu_2O and CuO) through endothermic decomposition of the nitrate. This process occurs at early stages in the reaction front.

Stage 2 (Post-Combustion Zone): The oxide phases are reduced to metallic Cu by reducing gases generated from fuel decomposition. This reduction occurs in the post-combustion zones, where temperatures remain elevated and reducing species are abundant.

The necessary condition for SCS of metals in the metal-nitrate–glycine system is that the oxidizer must decompose with formation of HNO_3 species . This decomposition pathway produces the NO_2/N_2O species that participate in the exothermic reaction with NH_3 from glycine decomposition, sustaining the combustion wave. For compositions with excess glycine, a hydrogen-reducing atmosphere develops in the reaction front, leading to reduced metal formation .

In Cu–Ni alloy synthesis, a fraction of the intermetallic Cu–Ni phase appeared directly in the combustion front, whereas the rest of the oxygen-free alloy formed through reduction of oxide phases . This suggests that the reduction kinetics depend on the specific metal and alloy composition.

3. SCS of Copper Nanostructures: Experimental Evidence

3.1 Copper Nitrate–Glycine System

The $Cu(NO_3)_2$–glycine system has been extensively studied as a model for metal SCS. Kumar et al. demonstrated that pure metallic Cu nanopowders can be obtained under fuel-rich conditions, with the product consisting of aggregated nanoparticles .

The phase evolution as a function of φ follows a predictable pattern:

φ Condition	Dominant Phase	Morphology
φ < 1 (fuel-lean)	CuO	Nanoparticles
φ = 1 (stoichiometric)	CuO + Cu_2O	Mixed
φ > 1 (fuel-rich)	Metallic Cu	Nanoparticles

The formation of metallic Cu requires φ sufficiently high to generate a reducing atmosphere, but excessive fuel can lead to residual carbon contamination and incomplete combustion .

3.2 Copper Nitrate–Citric Acid System

The $Cu(NO_3)_2$–citric acid system offers a contrasting case. Citric acid forms stable chelate complexes with Cu^{2+} ions, ensuring homogeneous mixing and promoting complete combustion . However, the higher reducing valence of citric acid (+18) compared to glycine (+10) means that the stoichiometric ratio requires less citric acid per mole of $Cu(NO_3)_2$.

Experimental studies on the $Cu(NO_3)_2$–citric acid system report the formation of **CuO nanorods** under certain conditions, with widths of 10–20 nm . The rod-like morphology is attributed to the anisotropic crystal structure of CuO and the combustion conditions that favor 1D growth. Critically, the product in this system is CuO, **not metallic Cu**. The combustion of citric acid with $Cu(NO_3)_2$ does not generate a sufficiently reducing atmosphere to produce metallic Cu, even under fuel-rich conditions, because the CO_2 and H_2O products dominate over reducing species.

3.3 Modified SCS Approaches for Metallic Copper

To overcome the limitations of conventional SCS for metallic Cu, researchers have developed modified approaches. Khort et al. reported a novel SCS modification using **microwave-assisted foam preparation** combined with **hexamethylenetetramine (HMT)** as fuel . This approach enables one-step synthesis of pure metallic Cu nanoparticles (~67 nm) under normal air atmosphere without post-reduction. The key innovation is the use of HMT, which decomposes to produce a strongly reducing atmosphere during combustion.

However, even with these modifications, the product consists of **spherical nanoparticles**, not 1D structures. The microwave-assisted foam preparation and rapid combustion process do not provide conditions favorable for anisotropic growth.

4. The Fundamental Limitation: Why SCS Cannot Produce Metal Nanowires

The analysis of SCS mechanisms reveals several fundamental obstacles to the direct synthesis of metallic Cu nanorods or nanowires:

4.1 Temporal Constraint

SCS combustion reactions complete within seconds to minutes, with the high-temperature reaction zone lasting only milliseconds to seconds . Anisotropic growth of nanowires requires sustained, directional deposition over timescales of minutes to hours. The combustion process is simply too rapid to allow for the selective facet adsorption and directional growth that characterize solution-phase nanowire synthesis .

4.2 Isotropic Driving Force

The driving force in SCS is the exothermic redox reaction, which proceeds uniformly throughout the reaction volume. This isotropic energy release promotes nucleation and growth in all directions equally, favoring spherical morphology to minimize surface energy. In contrast, solution-phase nanowire synthesis relies on **selective capping agents** (e.g., alkylamines, PVP) that adsorb preferentially on specific crystal facets, blocking growth in those directions and promoting 1D extension .

4.3 Gas Evolution and Particle Disruption

SCS produces large volumes of gaseous products (CO_2, H_2O, N_2, NO_x) that evolve rapidly during combustion . This gas evolution disrupts any incipient anisotropic structures and promotes the formation of porous, agglomerated powders rather than continuous 1D nanostructures.

4.4 Two-Stage Reduction Mechanism

The two-stage mechanism—oxide formation followed by reduction in the post-combustion zone—means that any 1D morphology formed during oxide growth would need to be preserved through the reduction step. Reduction of CuO to Cu involves significant volume changes and structural reorganization that would destroy 1D oxide morphologies .

5. SCS-Derived 1D Structures: Oxide Nanorods and Nanowires

While SCS cannot directly produce metallic Cu nanowires, it can produce 1D **oxide** nanostructures under carefully controlled conditions. The formation of CuO nanorods via SCS is well-documented . The mechanism involves:

1. **Chelate formation**: Citric acid forms stable complexes with Cu^{2+}, ensuring homogeneous mixing.

2. **Combustion**: The exothermic reaction produces CuO nuclei with anisotropic crystal structure.

3. **Growth**: The monoclinic CuO structure and combustion conditions favor growth along specific crystallographic directions, producing rod-like morphologies.

4. **Post-combustion**: The rapid cooling and gas evolution terminate growth, preserving the 1D morphology.

The aspect ratio of SCS-derived oxide nanorods is typically modest (length 100–250 nm, width 10–20 nm) compared to solution-phase Cu nanowires (lengths up to hundreds of micrometers, aspect ratios > 1,000) . This reflects the fundamental limitations of the combustion process for extended 1D growth.

Some reports describe the formation of 1D structures from SCS-derived nanoparticles through **post-annealing** . For example, $CaIn_2O_4$ nanorods were obtained by annealing SCS-derived nanoparticles, with self-assembly driven by the asymmetric crystal structure and high combustion temperature . However, this approach is not applicable to metallic Cu, which would oxidize during high-temperature annealing in air.

6. Solution-Phase Growth Mechanisms: A Comparative Perspective

To understand why SCS fails for metallic nanowires, it is instructive to examine the mechanisms operating in successful solution-phase Cu nanowire synthesis.

6.1 Selective Capping and Anisotropic Growth

Solution-phase Cu nanowire synthesis relies on capping agents—typically alkylamines (hexadecylamine, oleylamine, dodecylamine) or polymers (PVP)—that selectively adsorb on specific crystal facets . For Cu nanowires with fivefold twinned structure, capping agents adsorb on the {100} side facets, leaving the {111} end facets exposed for continued deposition . This selective passivation promotes growth along the ⟨110⟩ direction, producing nanowires with high aspect ratios.

Multiscale theoretical modeling has quantified this effect: density functional theory (DFT) calculations show that chlorinated Cu(111) surfaces bind Cu atoms more strongly and exhibit slower diffusion than chlorinated Cu(100) surfaces . The alkylamine layer covers most {100} facets but is absent or disordered on {111} facets, creating the differential growth rates necessary for 1D morphology .

6.2 Screw Dislocation-Driven Growth

A fundamentally different mechanism for Cu nanowire growth was identified by Meng and Jin: **axial screw dislocations** . Transmission electron microscopy (TEM) analysis conclusively characterized pure screw dislocations along the ⟨110⟩ direction, with associated Eshelby twist and spontaneously formed hollow nanotubes . This dislocation-driven growth mechanism provides a continuous step edge for anisotropic deposition and is characteristic of solution-phase synthesis, not combustion.

6.3 Supersaturation Control

Solution-phase synthesis allows precise control of supersaturation through reducing agent concentration, pH, and temperature . The reduction–oxidation electrochemical reaction forming Cu NWs can be tuned to promote dislocation-driven growth. SCS, by contrast, operates far from

equilibrium, with combustion occurring rapidly at high temperature and supersaturation that cannot be controlled in real time.

7. Conclusions

Solution combustion synthesis is a powerful technique for producing oxide nanomaterials, including CuO nanorods. However, its extension to **metallic** Cu nanorods and nanowires faces fundamental mechanistic obstacles:

1. **Thermodynamic constraints**: Metallic Cu formation requires fuel-rich conditions ($\varphi > 1$) to generate a reducing atmosphere, but these conditions favor isotropic nanoparticle formation rather than 1D growth.

2. **Kinetic constraints**: The combustion process is too rapid (seconds) and isotropic to support the sustained, directional growth required for nanowire formation.

3. **Mechanistic incompatibility**: SCS operates through a two-stage oxide-formation/reduction pathway that destroys any incipient 1D morphology, whereas solution-phase nanowire growth relies on selective capping and dislocation-driven mechanisms that have no analog in combustion synthesis.

SCS-derived 1D structures are currently limited to oxide nanorods (e.g., CuO) with modest aspect ratios. The direct synthesis of metallic Cu nanowires via SCS remains an unsolved challenge. Future work might explore hybrid approaches that combine SCS with post-synthesis templating or directed assembly, but the fundamental limitations identified here suggest that solution-phase routes will remain the primary method for metallic Cu nanowire synthesis.

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